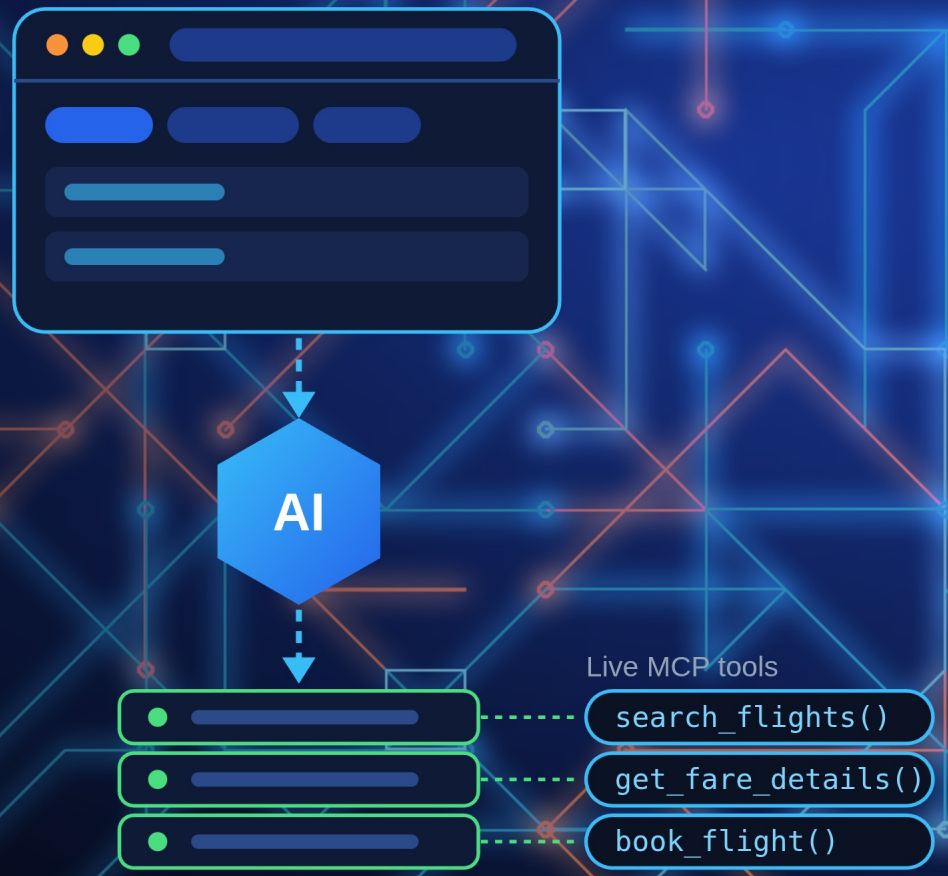
Kritical Hacks proposal using TinyFish

# Web2MCP

Turn any website into live, verified MCP tools for AI agents

Website → TinyFish + AI → MCP tools




TEAM

**6th Floor**

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PROBLEM STATEMENT

# Websites are built for humans, not agents

### No agent-friendly API

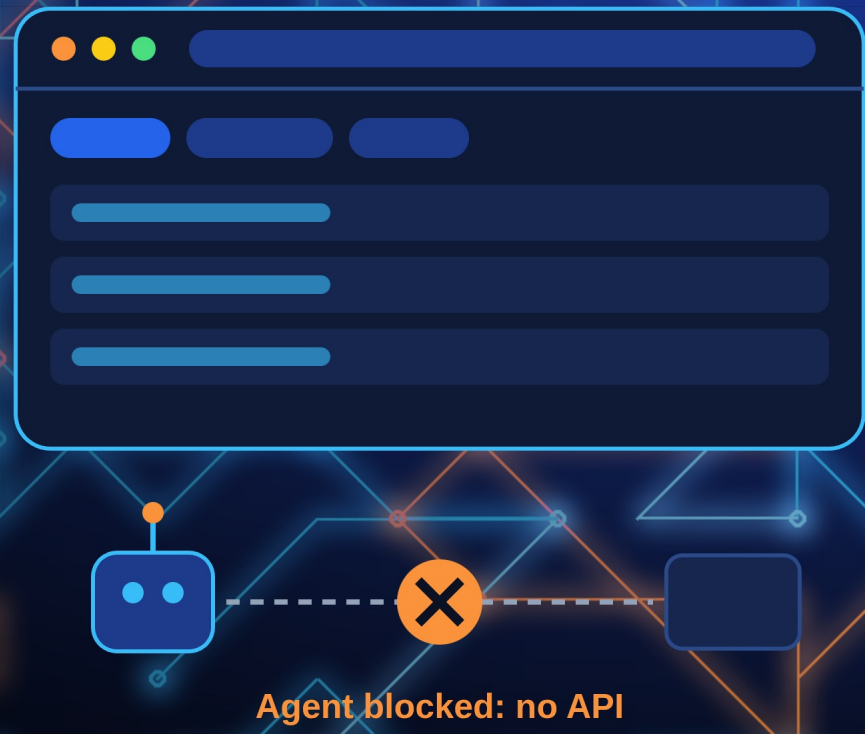
Most portals, listings and booking sites only work through a browser UI.

### Scrapers are brittle

A redesign or a renamed field silently breaks hand-written scripts.

### Every site is a new project

Forms, filters and pagination get rebuilt again and again for each site.



PROPOSED SOLUTION

# Give a website. Get live MCP tools.

### Point us at a site or goal
A URL or a plain task, such as a tender portal or travel site.

### TinyFish explores it
The agent opens pages, uses filters and forms, and reads results.

### AI infers and validates
Typed schemas are tested against real pages, with evidence kept.

### An MCP server is generated
Tools run in Claude, Cursor, VS Code and other MCP clients.

GENERATED MCP TOOLS

**search_flights(origin, destination, date)**

→ airline, fare, departure_time

✓ validated · evidence saved

**get_fare_details(flight_id)**

✓ validated · evidence saved

**book_flight(flight_id, passenger)**

⚠ requires user approval

TARGET USERS & USE CASES

# Built for people who connect agents to the web

| WHO | PAIN TODAY | WITH WEB2MCP |
| --- | --- | --- |
| **AI developers** | Wiring web actions into agents by hand | Ready MCP tools in minutes |
| **Agent builders** | Fragile scripts that fail mid-workflow | Typed, validated tools agents can trust |
| **SaaS companies** | No agent interface for their product | An official MCP server built from their site |

**Example sites: tender portals · careers pages · travel booking · property auctions**

ROLE OF MACHINE LEARNING / AI

# AI turns raw pages into trusted tools

### Understand the site

An LLM reads pages to find forms, filters, result lists and pagination.

### Infer the schema

Field names, types, enums and required inputs come from what was observed.

### Validate with evidence

Probe queries run live; extracted values are checked against the page and saved.

### Keep tools healthy

Scheduled re-checks flag drift and regenerate tools when a site changes.

**The hard part is inferring schemas and verifying behavior, not scraping pages**

PROPOSED TECHNICAL APPROACH

# From a URL to a reusable MCP server

### Input

Website URL or a plain-language goal

→

### TinyFish agent

Search, fetch and browser automation

→

### Extract and validate

Infer schema, check fields, keep evidence and errors

→

### MCP generator

Typed tool definitions and handlers

→

### MCP server

Reusable by Claude, Cursor, VS Code

BUILT WITH

**Node.js · TypeScript · MCP SDK · PostgreSQL · React dashboard**

SAFETY GATE

**Bookings, purchases and other consequential actions need user approval**

PROPOSED TINYFISH INTEGRATION

# TinyFish is the web acquisition layer

### Search
Discover the right portal and the pages that matter.

### Fetch
Pull clean page content for schema inference.

### Browser automation
Operate filters, forms and pagination to learn site behavior.

### Agent API
Tool handlers call TinyFish live for fresh results at runtime.

RUNTIME CALL

MCP client calls search_flights

↓

Web2MCP tool handler

↓

TinyFish agent drives the site

↓

Typed JSON and evidence returned

EXPECTED IMPACT

# The live web becomes agent-ready

## Build time

BEFORE

A custom scraper for every site

WITH WEB2MCP

**One request per site**

## Reliability

BEFORE

Silent breakage after site changes

WITH WEB2MCP

**Validated, monitored tools**

## Reach

BEFORE

One-off scripts for each agent

WITH WEB2MCP

**Any MCP client, reusable**

**Who benefits: agent developers, product teams and SaaS companies**

FEASIBILITY

# Achievable today with proven building blocks

## HACKATHON MVP

**Explore one public site**
TinyFish agent maps the pages and forms.

**Infer and validate 2–3 tools**
Checked against saved evidence.

**Generate the MCP server**
Typed definitions and handlers.

**Demo live in Claude**
Including the approval gate.

## RISKS AND MITIGATIONS

**Site redesigns**
Scheduled re-validation and regeneration

**Wrong schema**
Fields checked against saved evidence

**Login and bot walls**
Start with public pages, add sessions later

**Risky actions**
Explicit user approval before execution

FUTURE SCOPE

# Where Web2MCP grows next

## Now

- Single-site tool generation
- Validation with evidence
- User approval gate

## Next

- Auto-repair on site changes
- Authenticated sessions
- Tool versioning

## Later

- Public registry of verified tools
- Cross-site workflows
- Hosted remote MCP servers

**Vision: every useful website ships with a trusted MCP interface**

BUSINESS MODEL

# A usage-based platform for web tools

## Developer

**Free tier**

- Limited sites and tools
- Local MCP export
- Community support

## Teams

**Usage-based**

- More sites and tools
- Monitoring and auto-repair
- Shared workspace

## Enterprise

**Custom**

- Hosted official MCP servers
- SSO and audit logs
- Support and SLAs

**Revenue: generation and validation runs, hosting, enterprise plans**